

CAP-D'ESPOIR, QC, Canada

WMO: 714290

Lat: 48.419N

Lon: 64.317W

Elev: 51

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-1.5	-17.8	2.1	-4.9	-13.0	2.7	-1.0	40.7	27.9	35.6	28.7	10.1	270	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	70.2	63.6	68.1	63.2	66.4	62.3	66.0	68.3	64.6	66.8	63.2	65.4	10.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	92.6	67.4	63.6	88.0	66.0	62.2	83.8	64.7	30.8	68.5	29.8	67.0	28.8	65.4	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.5	27.9	24.2	DB	-11.0	77.1	4.4	3.9	-14.2	79.9	-16.7	82.2	-19.2	84.4	-22.4	87.2	(4)
			WB	-11.7	68.8	4.0	1.6	-14.6	70.0	-16.9	71.0	-19.1	71.9	-22.0	73.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	16.0	17.0	24.0	33.5	43.4	52.8	60.2	60.6	53.7	43.6	33.9	23.3
	DBStd	16.85	9.67	9.27	7.79	4.99	4.89	5.17	3.95	3.66	5.29	5.57	6.67	8.18
	HDD50	5063	1053	924	805	494	214	28	1	0	24	210	482	828
	HDD65	9637	1518	1344	1270	944	670	366	155	142	340	663	932	1293
	CDD50	915	0	0	0	0	9	112	317	330	135	12	0	0
	CDD65	15	0	0	0	0	0	0	7	7	1	0	0	0
	CDH74	11	0	0	0	0	0	1	4	4	2	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.6	11.5	11.4	12.1	11.2	10.2	9.2	8.1	8.6	10.1	11.3	11.7	12.3
Precipitation	PrecAvg	44.3	3.1	3.1	3.4	3.1	3.7	3.8	3.9	3.8	3.9	4.5	3.8	3.9
	PrecMax	57.3	5.1	5.9	6.9	6.1	6.9	7.5	6.2	6.2	6.4	9.1	7.6	8.5
	PrecMin	35.6	1.1	0.4	0.7	1.1	1.7	1.5	1.8	1.5	1.7	1.5	1.3	1.2
	PrecStd	5.0	1.2	1.3	1.5	1.5	1.5	1.3	1.1	1.3	1.3	1.7	1.3	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	37.6	41.2	49.3	62.3	70.3	73.7	73.8	69.9	61.1	50.2	41.5
		MCWB	37.7	35.4	35.5	40.6	52.7	59.6	64.3	64.8	62.0	53.7	47.2	39.6
	2%	DB	35.6	34.5	37.2	44.6	56.9	65.8	70.2	70.2	65.7	56.3	47.5	38.4
		MCWB	34.5	33.1	33.7	39.0	50.6	59.5	64.0	64.4	61.3	52.5	45.5	37.1
	5%	DB	32.8	32.2	35.0	41.8	53.4	62.9	68.2	68.1	63.1	53.8	45.3	36.3
		MCWB	31.8	31.3	32.6	37.7	48.4	58.1	63.4	63.7	59.8	51.2	43.5	35.0
	10%	DB	30.0	30.0	33.2	39.8	50.6	60.3	66.3	66.5	61.3	51.7	43.0	34.6
		MCWB	28.8	28.9	31.4	36.5	46.5	56.3	62.1	62.9	58.4	49.1	41.1	33.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	36.0	37.4	43.1	55.5	64.4	68.4	68.2	65.4	56.6	48.5	40.4
		MCDB	39.0	36.4	39.4	46.4	59.2	66.8	71.1	70.9	67.3	59.2	49.0	40.7
	2%	WB	34.4	33.3	34.7	40.4	52.0	61.3	66.0	66.2	62.6	53.7	45.9	37.3
		MCDB	35.2	34.2	36.2	42.2	55.4	63.8	68.5	68.4	64.2	55.2	46.8	37.9
	5%	WB	31.6	31.3	33.1	38.5	49.4	59.1	64.4	64.9	60.6	51.7	43.4	35.1
		MCDB	32.3	32.0	34.5	40.5	52.3	61.8	66.7	67.1	62.3	53.5	44.4	35.9
	10%	WB	28.6	29.1	31.6	36.9	47.1	57.0	63.0	63.7	58.9	49.4	41.2	33.3
		MCDB	29.4	29.9	33.0	39.1	49.9	59.7	65.3	65.8	60.9	51.4	42.3	34.2
Mean Daily Temperature Range	5% DB	MDBR	13.5	12.5	10.4	8.6	9.8	9.9	9.8	10.2	11.3	10.1	9.3	11.0
		MCDBR	15.8	10.8	10.8	11.8	14.9	13.9	12.7	12.4	12.9	12.4	10.2	10.3
		MCWBR	17.1	10.5	9.2	8.9	10.5	9.8	9.1	9.0	10.0	10.3	10.2	10.8
	5% WB	MCDBR	16.8	10.3	10.1	10.6	12.9	12.1	10.8	10.1	11.2	11.5	10.3	10.5
		MCWBR	17.4	10.4	9.3	8.7	10.3	9.8	8.7	8.3	9.5	10.5	10.4	11.1
Clear-Sky Solar Irradiance	taub	0.278	0.282	0.315	0.349	0.366	0.384	0.414	0.388	0.360	0.340	0.309	0.287	
	taud	2.380	2.300	2.300	2.389	2.423	2.374	2.338	2.404	2.467	2.494	2.495	2.422	
	Ebn at Noon	253	279	285	286	284	279	267	270	268	255	240	234	
	Edh at Noon	21	29	34	35	35	37	38	34	29	24	19	18	
All-Sky Solar Radiation	RadAvg	384	646	1051	1393	1636	1773	1714	1512	1155	713	375	275	
	RadStd	34	63	86	118	130	143	148	95	93	54	45	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)		N/A	N/A	+2.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page